PREPD – Dissertation Preparation

Mestrado em Engenharia Informática

Practical class 06

Ethics in engineering

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Overview

T1: Context

T2: Engineering ethics

T3: Codes of ethics for engineers and computing professionals

T4: Other aspects

Lecture 06, in the same week, covers ethics in research. This class covers ethics in engineering practice.

Before the class: play Survival of the Best Fit [9] (about six minutes, in the browser – next two slides). It is discussed in T4.

Before the class: Survival of the Best Fit

Survival of the Best Fit is an educational game about hiring bias in AI, built by Gabor Csapo, Jihyun Kim, Miha Klasinc and Alia ElKattan with a Creative Media Award from the Mozilla Foundation [9].

- You run a start-up that needs to hire quickly, and at some point you let a machine learning model screen the CVs.
- It shows, in a few minutes, how a system that meets its specification can still cause harm – the question of this class.
- It is not only a game: a real recruiting tool was abandoned in 2018 because it showed bias against women [10].

Before the class: Survival of the Best Fit – what to do

1. Open https://www.survivalofthebestfit.com/ in a browser and select "Start game" (about six minutes).
2. Play to the end, taking your own decisions.
3. Note down, to bring to the class:
   - at what moment did the system start to discriminate, and why?
   - who, in the game, could have prevented it – and how?
   - one requirement you would have written before automating the screening.

Discussed in T4 ("Activity – Survival of the Best Fit" and "Trustworthy AI").

T1: Context

T1: Master in Informatics Engineering – student outcomes

- ...
- An ability to recognize **ethical and professional responsibilities** in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
- ...

From https://www.isep.ipp.pt/Course/Course/87, English version.

T1: And engineers?

"With the Hippocratic Oath, doctors promise to work for humanity. Architects should promise the same. (en)"
"Com o Juramento de Hipócrates, os médicos prometem trabalhar para a humanidade. Os arquitetos deviam prometer o mesmo. (pt)"

Yasmeen Lari, architect, interviewed in [22].

- The term "engineering" derives from the Latin *ingeniosus*. In earlier times, few people were described as ingenious. Leonardo da Vinci (1452–1519), for instance, was referred to as engineer, architect or painter, according to the activity in question [25].
- In 1794, there were already schools dedicated to teaching engineering: "École Polytechnique was founded in 1794, under the name École Centrale des Travaux Publics, as a response to the dearth of engineers and high-level officials in France." It was renamed École Polytechnique in 1795 [33].
- Academic credentials were required of the members of France's first engineering association, established in 1848 [3].
- "Engineering is the application of an empirical, scientific approach to finding efficient, economic solutions to practical problems" [13].

Engenharia, uma profissão de confiança pública

– É objetivo da Engenharia preservar e melhorar as condições de vida da Humanidade.

Framed extract from [27].

T2: Engineering ethics

T2: Ethics

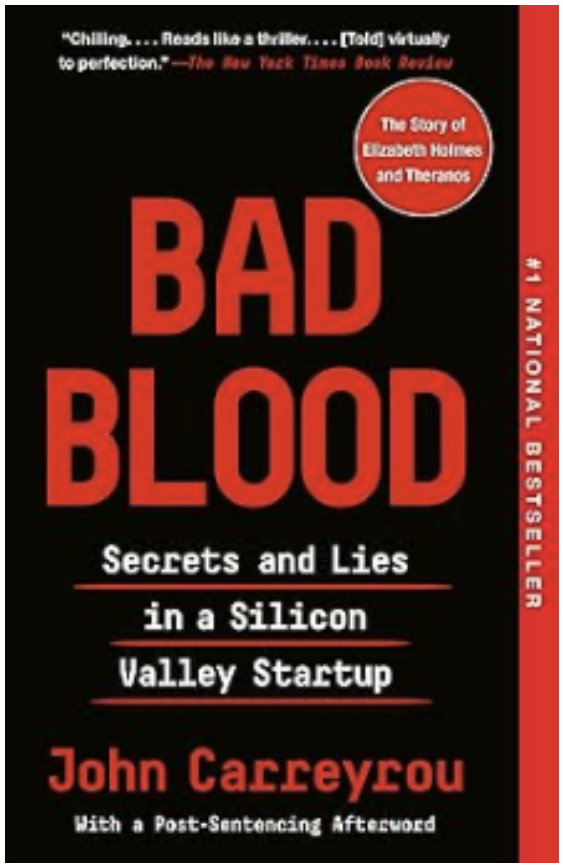
- "Ethics" and "morality" have twin roots: *ēthos*, in Greek, and *mores*, in Latin, both mean habit or character. However, a code of ethics is not regarded as a simple moral code [11].

- Fleddermann defines ethics as "the study of the characteristics of morals", which also deals with the moral choices that each person makes in his or her relationships with other persons [14].

- Engineering ethics is "the rules and standards governing the conduct of engineers in their role as professionals" [14]. One of the first books on the subject, *Ethical Problems in Engineering*, was published in 1965.

From [11, 21].

T2: Engineers' responsibilities (Theranos)



- Are the only legitimate interests of the employer, which the engineer has an ethical responsibility to serve, *purely technical interests*?
- Does an employer's legitimate interest in obtaining *the best technical work* from its engineers *involve identifying the organisational characteristics that impede or hinder it*?
- Strong managerial pressure on technical professionals to *work quickly*, in order to get an innovative product to market as soon as possible, often puts engineers in an ethical dilemma. This pressure can lead to shortcuts and irresponsible behaviour.

Based on [24].

T2: Engineers' responsibilities (Theranos) (2)

- "Having worked at Apple and Adobe as a product designer and manager, Ana Arriola became Theranos's chief design architect in August 2005"
- "Arriola feared the human subjects might be used as guinea pigs to test an unreliable device, something she believed would cross an ethical line."
- "An ethical issue that Arriola faced was whether to continue working for **a company she believed was prioritizing profit over not putting human patients at risk** with its work, even if she was not part of the group conducting the pilot study."
- Arriola left the company.

From [24].

T2: Engineers' responsibilities (Theranos) (3)

- "Engineer Aaron Moore [...] was concerned about the effects of what he viewed as **poor communication** between Theranos's engineering and chemistry groups, not just organizationally but in terms of the downstream **consequences of such communication for humans**."

- He left the company.

From [24].

T2: The need for ethical guidelines

- "The radar screens at IGI's ATC went blank at 5.50 pm after the Autotrac III software system, on validation trial, developed a snag and crashed... The system was restored at 6.14 pm. by switching over to the previous Autotrac II system... More than 20 flights were directly affected during this period and about 50 flights were delayed by the backlog that followed the malfunction." [2] – and it was not the last failure.

- "Toyota has admitted that most 2010 Prius models (those built between the start of production last spring and late January) have a glitch in the software that controls the anti-lock braking system." [15]

- The Boeing 737 MAX disasters [17].

- Therac-25: radiation overdoses [23].

- "Volvo has a problem with brake software. In Portugal, there are 522 cars affected" [30].

T2: Boeing 737 MAX disasters

- **Context:** The Manoeuvring Characteristics Augmentation System (MCAS) automatically pushed the nose of the aircraft down to compensate for the effect of its larger, repositioned engines. It acted on data from a single angle-of-attack (AoA) sensor. Two accidents, in October 2018 and March 2019, killed 346 people.

- **The debate:** Was this a software problem? The code did what the specification said.

See [17].

T2: Boeing 737 MAX disasters (2)

The engineering failure (requirements)

"Nowhere in here is there a software problem. The computers & software performed their jobs according to spec without error. The specification was just ######."
https://twitter.com/trevorsumner/status/1106934422249582593

- **Reframing:** Not specifying redundant sensor inputs (e.g. the use of the two AoA sensors) for a safety-critical function is a failure of requirements engineering and of safety analysis.
- **Conclusion:** The code was not wrong, but the software engineering process failed.

Ethical reading

The disasters conflict with several ethical guidelines, including:

- **Public (SE Code, 1.03):** approve software only with a well-founded belief that it is safe; the NSPE code asks engineers to hold paramount the safety, health and welfare of the public.
- **Client and employer (SE Code, 2.09):** did the pressure to cut costs and to avoid expensive pilot retraining (the interest of the employer) compromise the "higher ethical concern" of public safety?

See also [31] and the Software Engineering Code of Ethics and Professional Practice.

T2: The Therac-25 tragedy

- **Context:** Between June 1985 and January 1987, the Therac-25 radiation therapy machine was involved in six known accidents in which patients received massive overdoses, resulting in deaths and serious injuries.

- **The core issue:** Software was reused from earlier machines, without adequate testing or safety review in the new system. Reused software that has worked without known problems still needs proper reviews and tests.

See [23].

T2: The Therac-25 tragedy (2)

The technical failure

The accidents were caused by a combination of factors:

- **Race condition:** a specific sequence of inputs (the operator editing the treatment data quickly) led the software to bypass safety checks.
- **Design flaw:** safety relied exclusively on software. The independent protective circuits and mechanical interlocks of the earlier Therac-20 (physical mechanisms that prevented unsafe operation) had not all been kept.
- ...

Ethical reading

A failure of professional responsibility:

1. **Inadequate verification:** there was no adequate record of the tests made before the software went into production.
2. **Public safety:** the paramount obligation to protect the health and welfare of the public was not met.
3. ...

T3: Codes of ethics for engineers and computing professionals

T3: Code of ethics

A code of ethics provides a "framework for ethical judgment" [14]. Common designations are code of conduct, code of ethics or, simply, guidelines. Some examples:

- Code of Good Practices and Conduct of P.PORTO (Despacho n.ž 11171/2020) [20];
- Code of Ethics and Deontology of the Ordem dos Engenheiros, the Portuguese professional association of engineers [27].

T3: First codes of ethics for engineers

- Code of Principles of Professional Conduct of the American Institute of Electrical Engineers (AIEE): 8 March 1912.
- Ethical Guidelines for Norwegian engineers: 1921.
- Ethical guidelines adopted by the Chinese Institute of Engineers: 1933.

From [11, 21].

T3: Commonalities in engineering codes

- do not. . .

> O engenheiro não deve aceitar trabalhos ou exercer funções que ultrapassem a sua competência ou exijam mais tempo do que aquele de que disponha.

- must. . .

> Os engenheiros devem usar de prudência, tanto nas soluções técnicas, como no desenvolvimento tecnológico e na inovação, responsabilizando-se pela qualidade, fiabilidade e segurança dos produtos e processos técnicos concebidos ou executados, promovendo a criação de valor económico e social de forma sustentável.

- common good

> O engenheiro não deve divulgar nem utilizar segredos profissionais ou informações, em especial as científicas a técnicas obtidas confidencialmente no exercício das suas funções, salvo se, em consciência, considerar poderem estar em sério risco exigências do bem comum.

- law

> Os engenheiros devem conhecer as leis e as regulamentações relevantes dos países onde exercem a profissão e respeitá-las, desde que não contradigam princípios éticos universais, comprometendo-se a aplicar, com espírito crítico, as suas competências profissionais, visando a melhoria daquelas.

- engineers should maintain and improve their technical expertise: competence is usually part of what the codes require.

Framed extracts from [27].
Competence: "the ability to apply knowledge and skills at a certain level of independency and autonomy" [8].

T3: Three codes of ethics for engineers and computing professionals (3CEECP)

- National Society of Professional Engineers (NSPE) Code of Ethics for Engineers
- ACM Code of Ethics and Professional Conduct, for computing professionals
- ACM/IEEE-CS Software Engineering Code of Ethics and Professional Practice

All these codes mention:

- human well-being and the public interest;
- the need to be honest and to avoid deception;
- competence;
- privacy and confidentiality.

T3: Primary scope (3CEECP)

ACM Code of Ethics	**ACM/IEEE-CS Software Engineering Code**	**NSPE Code of Ethics**
"all computing professionals, including current and aspiring practitioners, instructors, students, influencers, and anyone who uses computing technology in an impactful way."	Professional software engineers, "including practitioners, educators, managers, supervisors and policy makers, as well as trainees and students of the profession".	Engineers of all disciplines, "in the fulfillment of their professional duties".

T3: Structure (3CEECP)

ACM Code of Ethics	**ACM/IEEE-CS Software Engineering Code**	**NSPE Code of Ethics**
Four sections: 1. General Ethical Principles 2. Professional Responsibilities 3. Professional Leadership Principles 4. Compliance with the Code	Eight principles: 1. Public 2. Client and Employer 3. Product 4. Judgment 5. Management 6. Profession 7. Colleagues 8. Self	Three main parts: 1. Fundamental Canons 2. Rules of Practice 3. Professional Obligations

T3: Avoidance of harm (3CEECP)

ACM Code of Ethics	**ACM/IEEE-CS Software Engineering Code**	**NSPE Code of Ethics**
Explicit ("Avoid harm"): it addresses bias, misuse and unintended consequences.	Through product quality, testing and verification, and responsible judgement.	Through safety, competence and truthful communication, to prevent risks.

T3: Recommendation

Choose one engineering code of ethics and follow it – but not selectively.

T3: A method for ethical decisions

A code says what matters; a method helps to decide in a concrete case [32]:

1. **Identify the ethical issues**: could the decision harm someone, or benefit some people unevenly?
2. **Get the facts**: what is known, what is not known, who has a stake?
3. **Evaluate alternative actions**, through several lenses: rights, justice, utilitarian (consequences), common good, virtue and care ethics.
4. **Choose an option and test it**: which option best addresses the situation? Could you explain it to the people affected?
5. **Implement the decision and reflect on the outcome**: what did you learn?

More material: Ethics in Technology Practice, from the same centre.

T4: Other aspects

T4: Ethical requirements

- In software engineering, **ethical requirements** must be captured and analysed with the stakeholders, including users and developers, to promote ethical alignment in software artefacts and development processes [7]. The authors discuss the need to "identify use and mis-use cases and translate ethical concerns into concrete system behavior".

- And in other areas?

T4: Activity – Survival of the Best Fit

In Survival of the Best Fit [9], you hire by hand and then automate the screening with a model trained on your own past decisions.

- Where did the bias of the model come from?
- Who was responsible for the discrimination: the CEO, the engineers, the investors – or "the data"? Can data be responsible?
- Which **ethical requirement** was missing, and how could it have been tested before the system was used?

Not only a game: in 2018, Amazon abandoned an AI recruiting tool that showed bias against women [10]. Under the AI Act, AI systems used for recruitment and selection are classified as **high-risk**.

T4: Trustworthy AI

The seven requirements of the Ethics Guidelines for Trustworthy AI [18]:

1. human agency and oversight;
2. technical robustness and safety;
3. privacy and data governance;
4. transparency;
5. diversity, non-discrimination and fairness;
6. societal and environmental well-being;
7. accountability.

Which of them failed in Survival of the Best Fit? If your dissertation uses machine learning, which of them apply to it?

T4: Possible ethical dilemmas

- Is it unethical not to collect requirements from key stakeholders or users of a system? Recommendation: "include considerations of justified stakeholder selection" [28].
- Who should be responsible for adding higher levels of scrutiny when a system that was not safety-critical evolves into a safety-critical one?
- Who is responsible for the unknown ethical impacts of algorithms?
- If an application is found to be addictive and detrimental to its users, should the ethical practitioner develop techniques to reduce dependence, or choose to sunset the application and avoid involvement in similar applications?

From [16].

T4: Ethical questions on Stack Exchange (Q&A) websites

- If the spec is flawed, should it still be followed? ("Does a programmer have an ethical responsibility NOT to implement code with known security risks?")
- What to do if you find a vulnerability in a competitor's site? ("...would you report a serious vulnerability to your direct competition, in order to help them?")
- Is it better to blindly match the project requirements or to match the spirit of the project requirements?
- Should you promise to deliver a feature that you aren't sure if it is implementable?

T4: Bad quality / technical debt

- **Context:** The software passes all the tests and the user is happy (high external quality), but the code is of poor internal quality.
- **The common question:** "Is it unethical to present poor quality code that violates coding standards?"

Adapted from [4] and the ACM Code of Ethics.

T4: Bad quality / technical debt (2)

The ethical conflict

- **Technical debt:** choosing a fast, tangled solution that does not conform to the standards creates technical debt.
- **Future harm:** the debt must be paid by someone – usually the engineers who maintain the code, or the client, who will face higher costs later.

Professional responsibility

The arguments most often given for regarding this as an ethical problem:

1. **Competence:** it can violate the professional duty to produce work to the best of one's ability.
2. **Integrity:** it misrepresents the true long-term robustness and cost of the software to the client or employer, unless the debt is disclosed.
3. **Colleagues:** it pushes the maintenance effort and the rework onto other engineers.

Adapted from [4] and the ACM Code of Ethics.

T4: The Maven groupId dilemma

- **Context:** In Java/Maven, the `groupId` (e.g. `com.google.guava`) acts as a namespace that identifies the **organisation or project that publishes** the artefact.
- **The question:** "Is it okay to change the open-sourced Maven `groupId`?" [4]

T4: The Maven groupId dilemma (2)

The ethical conflict

- **Risk of impersonation:** if you keep the original `groupId` but publish your own version, you are presenting yourself as the original publisher.
- **Risk of plagiarism:** if you change the `groupId` to your own (e.g. `com.mycompany`), it may look as if you are claiming credit for the entire codebase.

Common resolution

If you must publish a forked version (e.g. the original has been abandoned):

1. change the `groupId` to your own (a technical necessity: a publisher can only use a namespace it controls);
2. keep all the original licence files and copyright headers, and state clearly that it is a fork, so that the original work is properly attributed.

T4: Forking etiquette

- **Context:** Open-source licences (such as MIT or GPL) legally allow you to copy and modify code. However, community norms are stricter than the law.
- **The question:** "What are the ethics & etiquette of forking someone else's project to release it as a package (Gem/NPM)?" [4]

T4: Forking etiquette (2)

The inappropriate fork

Releasing a competing version of a library without contacting the original author is considered poor etiquette. It splits the community and duplicates effort.

Professional norms (the order of operations)

1. **Contribute first:** submit a pull request (PR) to the original repository.
2. **Wait:** give the maintainer time to respond.
3. **Justify:** only release a separate fork if the maintainer is unresponsive, has abandoned the project, or explicitly rejects a feature you critically need.

T4: Open-source licences

- Many questions on Stack Exchange (Q&A) websites are about the use of open-source software [4].
- The incorrect use of components (e.g. including GPL code in a closed-source commercial plugin that is distributed) can amount to a licence violation, and therefore to copyright infringement.

T4: Open-source licences (2)

Category	Common licences	Key obligations
Permissive (minimal restrictions)	MIT, Apache 2.0, BSD	Allows reuse in proprietary software. You must keep the original copyright notice and the licence text. *Usually compatible with commercial projects.*
Weak copyleft (file- or library-level)	LGPL, Mozilla Public License (MPL)	You can use it from proprietary code, but if you modify *the library itself* (LGPL) or *its files* (MPL), you must share those modifications.
Strong copyleft ("viral")	GPL v2, GPL v3, AGPL	If you distribute software that incorporates it, the **entire** source code (including your own work) must be made available under the same licence. With the AGPL, this also applies when the software is offered over a network.

T4: Ethics and law

"Ethics is both more and less than law: it is more because many ethical concerns are not addressed by the law and less because the outcome of ethical considerations is not necessarily transformed into legal norms and thus not enforceable by way of law." [19]

See also lecture 06:

Most societies also have legal rules that govern behaviour, but **ethical norms tend to be broader and more informal than laws**. Although most societies use laws to enforce widely accepted moral standards, and ethical and legal rules use similar concepts, ethics and law are not the same. An action may be legal but unethical, or illegal but ethical.

From [29].

T4: Law

- Cyber Resilience Act – Regulation (EU) 2024/2847 of the European Parliament and of the Council of 23 October 2024 on horizontal cybersecurity requirements for products with digital elements ("From baby-monitors to smart-watches, products and software that contain a digital component are omnipresent in our daily lives...")
- AI Act – Regulation (EU) 2024/1689
- Copyright in cyberspace
  - EU copyright law, including the Software Directive (Directive 2009/24/EC)
  - open source and free access
- Regulation of distributed ledger technologies (DLTs) and smart contracts, e.g. MiCA (Regulation (EU) 2023/1114) and the Data Act (Regulation (EU) 2023/2854)
- EU data protection law – the GDPR, Regulation (EU) 2016/679

T4: Codes of ethics and law

- Many codes of ethics refer to the law and to the need to consider "existing rules pertaining to professional work", for instance the ACM Code of Ethics and Professional Conduct (principle 2.3).

- **Know and respect existing rules pertaining to professional work**: "Rules" here include local, regional, national, and international laws and regulations, as well as any policies and procedures of the organizations to which the professional belongs. Computing professionals must abide by these rules unless there is a compelling ethical justification to do otherwise. **Rules that are judged unethical should be challenged.** A rule may be unethical when it has an inadequate moral basis or causes recognizable harm. A computing professional should consider challenging the rule through existing channels before violating the rule. **A computing professional who decides to violate a rule because it is unethical, or for any other reason, must consider potential consequences and accept responsibility for that action.** (Quoted from the code.)

T4: Whistleblowing

- A **whistleblower** is "a person who reports suspected illegal or unethical activity" [1]. Boisjoly (Challenger, lecture 06), Arriola and Moore (Theranos) faced this decision.
- The codes expect it: the Software Engineering Code asks engineers to disclose to appropriate persons or authorities any danger to the public associated with software (clause 1.04), and to inform them when a higher ethical concern is being compromised (clause 2.09).
- The law protects it: Directive (EU) 2019/1937 [12], transposed in Portugal by Lei 93/2021 [5], protects people who report breaches learnt in a professional context – interns included – and prohibits retaliation.

Report through the internal channels first, when they exist and are safe to use.

T4: Recommendation (again)

Choose one engineering code of ethics and follow it – but not selectively.

Closing

Four things to carry forward:

- the responsibility of an engineer is not purely technical: **the safety, health and welfare of the public come first**, even under pressure from the employer or from the schedule;
- software that meets its specification can still cause harm – **requirements, safety analysis and testing are where ethics becomes engineering** (737 MAX, Therac-25);
- **legal is not the same as ethical**: licences, regulations and codes of ethics all apply to the dissertation work, and a code should be followed as a whole, not selectively;
- a hard case is decided **with a method and with the code in hand** – and those who raise a concern are protected by law.

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